



Fig. 5: Block diagram of VLC receiver

TABLE II
VLC VERSUS RF

Property	VLC	RF
Visibility	Yes	No
Frequency	400THZ-790THZ	850nm-950nm
Security	More	Less (compared to VLC)
Power consumption	Relative low	Medium
Coverage distance	Short	Medium
Harmful for human body	No	Yes

for faster transmission of data. The signal produced by the transmitter is received completely by the receiver. The surrounding illuminating light increases the base luminance but does not alter the variations produced by the transmitter. By adopting the base luminance of an environment, variation produced can be fully received by the receiver even after reflection. The most commonly used photoconductive cell is LDR. Silicon photodiode, phototransistor, avalanche photodiode, PIN diode and camera (array of light-sensitive elements) can also be used as a receiver.

The coding technique (OOK modulation) used in the transmitter section is interrupted and the corresponding decoding technique is used. Decoding is done by the MCU or microprocessor at the receiver end. Binary representation of each character is stored at the receiver end, too. Here, it is used for matching the obtained signal via LDR from the LED with actual binary representation. For example, 7-bit code for character A is assumed to be [011 0001] in transmitter section. It is saved at the receiver end so that it can compare the obtained signal word with the actual word.

Delay time for the receiver is

matched to delay time of the transmitter. The synchronised process takes care that the information sent by the transmitter is received fully at the receiver end. A small delay between transmitter and receiver end can cause an error.

Variations in LDR due to light emitted by the LED at the transmitter end are continuously observed by Arduino. Received variation is amplified and compared with the binary representation of the character, which is predefined in the transmitter. Thus, the character is displayed as output in the computer via Arduino's serial monitor.

Characteristics of VLC

Frequency. Visible light has higher frequency than radio waves, and spectrum for visible light is not limited. Visible light frequencies lie between ultraviolet (UV) and IR waves. Visible light is divided into four ranges:

1. Near UV (3PHz)
2. Far IR (3THz)
3. Mid IR (30THz)
4. Near IR (300THz)

Speed. Speed of VLC can be up to 100 times the speed of Wi-Fi. In laboratories, speed of Li-Fi can be up to 224Gbps using suitable modulation techniques.

Power. VLC does not require external power as it has power for LED light. It simply alters the light emitting frequency to transmit data. Illumination lights are used in this technology.

Antenna. The antenna frontend is not required in this technology, which reduces power consumption greatly. Since existing illumination LEDs are used, a transmitter is not required.

Similarly, the receiver is a photo-sensitive element. There is no restriction in the angle of projection. Hence, more receivers can benefit from a single transmitter.

Security. VLC uses visible light for communication. Hence, it is easy

to determine who receives the message. Also, it is impossible to tap the communication without breaking the link.

Human safety. VLC does not affect the human body at all. Thus, transmission power can be kept high if needed.

Bandwidth. VLC has a bandwidth range from 430THz to 750THz. This range is larger than the bandwidth in RF communications, which is from-3kHz to 300GHz.

Unlicensed spectrum. No company owns property rights for visible light and, thus, no royalty fees have to be paid. No expensive patent/licence needs to be purchased to use visible light for communication purposes.

Applications of VLC

Smart lighting. Smart buildings require smart lighting. Smart lighting with VLC technology provides better infrastructure for illumination, control and communication. Since communication between LEDs is possible, additional wirings are reduced and, thus, power consumption is greatly reduced.

The buildings can be designed to function in many ways. For example, if VLC technique is used in Burj Khalifa (currently the world's tallest building), it can either illuminate the whole building in a particular pattern or it can transmit the message over a great distance, making communication among its neighbours possible.

Mobile connectivity. By pointing visible light from a Li-Fi modem to another device, a high-speed data link with inherent security can be created. This overcomes the problem of pairing or connecting, by providing a much higher data rate than Bluetooth or Wi-Fi.

In hazardous environmental areas where there is a risk of explosions (mines, petro-chemical plants and oil rigs) and radioactivity-free zones (hospitals, reserved forests, etc), VLC is inherently safe and provides both illumination and communication.

Vehicles and transportation. Traffic signage, traffic lights and street lamps are adopting LED technology, so there are multiple application